Tensile Workbench — Reply to Audit Response

Follow-up to the review of the independent audit (commit fdcb232) · 26 October 2026

The response's qualifications are mostly right. This reply answers its four requests: reproduction materials, separation of user-facing failures from isolated concerns, the qualified remedies, and standards references. The most important new result is that **C3 is more severe than the audit reported**.

No application code was changed. Commit 4bd8b8a could not be checked because the public repository only contains fdcb232; running the reproduction bundle on that commit will show which findings still hold.

1. Fixtures, seeds and commands

The accompanying archive **tensile_audit_repro.zip** rebuilds every numerical example from fixed generators and seeds. Each check prints **DEFECT PRESENT** or **not reproduced**, so the same scripts can later serve as closure evidence.

- Run everything: `TW_REPO=/path/to/checkout python checks.py`; selected checks: `python checks.py C3 C4 C7`; C3 sweep: `python c3_sweep.py`.
- Safety: writes only to new temporary folders, sets `TENSILE_WORKBENCH_BACKUP_DIR` to a temporary folder before importing the application, and never reads or writes `data/`, `output/` or personal settings.
- Reference outputs for fdcb232 are included (`results_fdcb232.json`, `sweep_fdcb232.txt`); every C1–C10 number in the audit report reproduced exactly.
- Environment: Python 3.12.3, numpy 2.5.4, pandas 3.1.0, anywidget 0.11.0.

Generators

Generator	Used by	Definition
ramberg_osgood	all curves	Inverts e = s/E + 0.002·(s/ys)^(1/n) on 300 000 stress points; Rp0.2 equals ys exactly by construction.
necking_curve	C4, C5, C6, abrupt control	Ramberg–Osgood to eu, quadratic 20 % stress decline to ef, drop to zero over 3 readings; 0.1 s per reading; no noise. C4: E = 200 GPa, ys = 250 MPa, eu = 25 %, ef = 35 %, 15 000 readings, n = 0.05–0.40.
progressive_curve	C3	Ramberg–Osgood to 18 %, then su·(1 − t^p) falling smoothly to zero at 30 %; no sudden drop. Noise from numpy default_rng(seed), then rounding to quantum q, then clipping at 0. Defaults 8000 readings, seed 1. Reference endpoint: reading before the first post-peak load below 10 % of peak.
fixture_coupon	session checks	Same shape as tests/selection_fixture.py.

2. C3 correction: worse than reported

The audit's C3 table used one noise seed. A sweep over 20 seeds on smooth progressive failures that contain no sudden drop:

Signal	Mislabelled "sudden drop"	Worst EL error
Noise 0.05 MPa	20/20	−5.95 pp
Noise 0.1 MPa	20/20	−4.16 pp
Noise 0.2 MPa, 4000 readings	19/20	−4.03 pp
Noise 0.2 MPa, 8000 readings	8/20	−2.41 pp
Noise 0.2 MPa, 16 000 readings	1/20	−1.55 pp
Noise 0.5 or 1.0 MPa	0/20	—

Signal	Mislabelled “sudden drop”	Worst EL error
Quantised 0.01 MPa + noise 0.02	20/20	−9.99 pp
Quantised 0.05 MPa + noise 0.02	13/20	−8.37 pp
Quantised 0.1 MPa + noise 0.02	0/20	—
Quantised 0.5 MPa + noise 0.02	20/20	−1.02 pp
Quantised 1.0 MPa + noise 0.02	20/20	−2.93 pp
Accelerating tail p = 4, noise 0.2 MPa	20/20	−2.72 pp
Accelerating tail p = 8, noise 0.2 MPa	20/20	−1.12 pp

In the worst case, a 0.13 MPa decrement at 98.7 % of peak load (preceded by a 0.02 MPa change, ratio 6.5) was labelled an ISO-confirmed sudden drop with no review flag; reported EL was 19.39 % against a correct 29.38 %.

This explains why the response’s noisy probe did not misclassify: the failure depends on noise *relative to* the local decrement and on sampling density, and it is not monotonic. It supports the response’s point that any replacement threshold must be validated across a grid, not tuned on one example. A genuine abrupt fracture (control) is still detected correctly.

3. Which findings are user-facing

Evidence level	Meaning	Findings
Full session	PreviewSession — the object the UI calls for loading, matching, tables, averaging and Excel export; no browser	C1, C2, C7, C9, C10, landmark control
Application functions	The app’s loader and calculation functions on CSV files written to disk	C3, C4, C5, C6, C8, “Time (s)” header
Isolated	Widget or function outside its UI wiring; no reachable UI path shown	Variable “Save all”, representative override
Code reading only	Not executed	Cloud-sync behaviour

- **Withdrawn — pointwise “ddof = 0” SD.** Both callers discard that value (`tensile_core.py:1183` and `:1637`), so it never reaches a plot or export.
- **Confirmed — “Time (s)” header.** With no units row, all 3003 time values became NaN, so gap detection silently falls back to row order.

4. Qualified remedies

Item	Position	Detail
Tested counts	Agree	Deriving “tested” from what loads was wrong. Separate counts for tests found, curves loaded and imports needing attention are better.
Location bounds	Agree	Nothing should be rejected. Show which label token supplied the default and flag ambiguity; treat a value above the count as a soft notice only, since build-wide numbering is legitimate.
Two preparation paths	Agree, with clarification	Fracture detection must keep acquisition order. The concern is narrower: the two sorted views disagree — one drops strain-reversal rows, the other keeps them — so UTS/UE and EL/toughness come from different point sets. Shared validation plus flagged reversals is the right fix.
Excel formatting	Conceded	Strikethrough does not stop AVERAGE(). Eligible values in the main columns and raw values in labelled columns is the right remedy.
Backup retention	Conceded	“Add time-based retention” was poorly worded: 30 daily checkpoints already exist. The real gap is recovery within a single day; bounded hourly checkpoints would address it.

5. Standards

Verified through secondary sources and older editions

- **ASTM E8/E8M §7.11.3** — a published Standards Update Notice for E8/E8M-16a quotes the rule: for materials without a sudden decrease in force, elongation at fracture is the strain measured just before force falls below 10 % of maximum. Source: Intertek SUN, ASTM E8/E8M.
- **ASTM E8-03 §3.1.5** — uniform elongation includes both elastic and plastic elongation. Source: E8-03 copy, University of Michigan.
- **ISO 6892-1:2019 terms** — A is permanent elongation after fracture; At is total (elastic + plastic) elongation at the moment of fracture; Agt and Ag are total and plastic elongation at maximum force. Source: a test report quoting EN ISO 6892-1:2019 definitions. Annex A (computer-controlled testing) is informative, per the ISO preview contents.

Interpretation

Calc EL is an At-like total strain at the selected endpoint, not A. It should be labelled that way in plots and exports.

Still outstanding

- The ISO 6892-1:2019 A.3.6 wording behind the >5× / <2 % rule — no reliable quotation was found.
- ASTM E8/E8M-25 clause numbering (§7.11.3.3 and §7.11.3.4) and any change in wording since the 2016 edition.

Both need the licensed full texts before the documentation cites them.